Transformations, Projection & Curves from Paintings

CS405 Introduction to Computer Graphics · Fall 2026 · Project 1 report

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1 · Painting and scene

Painting: *Kavunlu Natürmort* (Still Life with Melon) by **Şeker Ahmed Paşa**, Sakıp Sabancı Museum collection. I chose it for the curve part. The whole melon has one long, clean contour against a dark background: the cut face on the left, the arc of the rind across the top and the rounded right end. That makes it a good shape to trace with a few cubic segments. The composition is also simple, a single object on a table, so a 3D table in front of the canvas continues the painting naturally.

The painting hangs as a textured quad in the $z = 0$ plane, inside a wooden frame and against a wall. In front of it is a table. On the table are the Part A box, a melon on a plate (a nod to the painting) and a desk lamp whose arm moves (Part B). The traced outline and a small cone that travels along it (Part D) sit on the painting itself. Everything is one WebGPU page (Figure 1).

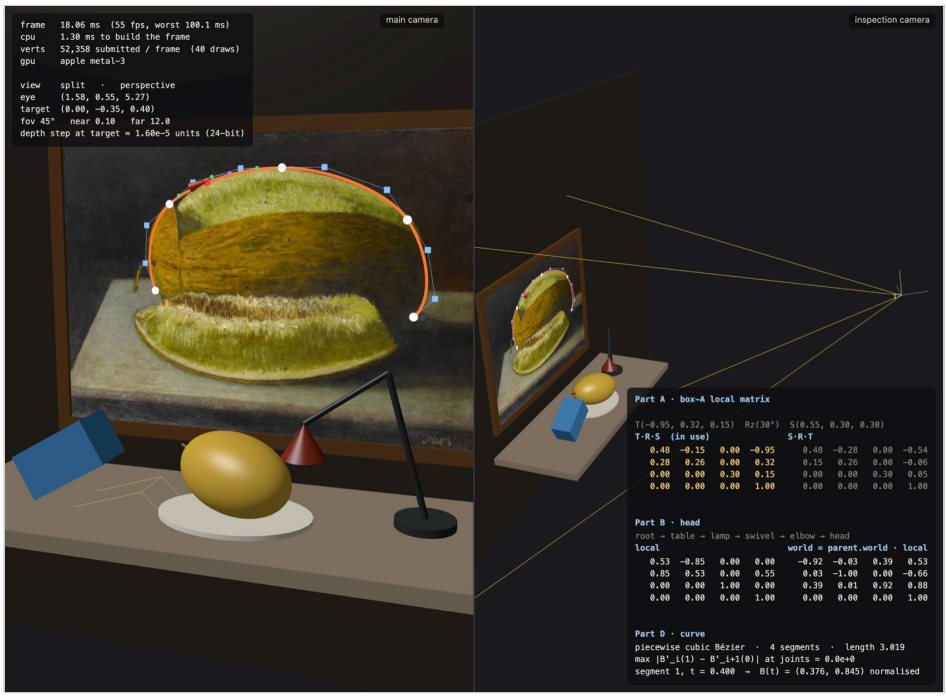


Figure 1. The application in side-by-side mode. Left: main (orbit) camera with the traced melon outline, control polygon and the de Casteljau points. Right: fixed inspection camera showing the main camera's frustum. Panels show timing, camera parameters and the matrices of Part A, B and D.

2 · Part A: transformations

Convention. I use column vectors, $p' = M\,p$, and a right-handed world with y up. Matrices are `Float32Array(16)` in column-major order: entry (r, c) is stored at index $4c + r$. This is exactly how WGSL reads a `mat4x4f` from a uniform buffer, so nothing is transposed on upload. Because vectors are multiplied on the right, $M = T{\cdot}R{\cdot}S$ applies the scale first and the translation last. `src/math/mat4.js` implements identity, multiply, translate, rotate about x/y/z, scale, transpose and the inverse of a rigid transform, $M^{-1} = [R^{\mathrm{T}} \mid -R^{\mathrm{T}}t]$. The rigid inverse is used again for the view matrix and for converting mouse rays into the painting's frame.

For the box with $T(-0.95, 0.32, 0.15)$, $R_z(30°)$ and $S(0.55, 0.30, 0.30)$ the two orders give:

$$T{\cdot}R{\cdot}S = \begin{pmatrix} 0.476 & -0.150 & 0 & -0.950 \\ 0.275 & 0.260 & 0 & 0.320 \\ 0 & 0 & 0.300 & 0.150 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad S{\cdot}R{\cdot}T = \begin{pmatrix} 0.476 & -0.275 & 0 & -0.540 \\ 0.150 & 0.260 & 0 & -0.059 \\ 0 & 0 & 0.300 & 0.045 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In $T{\cdot}R{\cdot}S$ the box is scaled in its own axes, rotated, then moved. The last column is exactly the translation I typed, and the columns of the 3×3 block stay perpendicular, so the box keeps its right angles. In $S{\cdot}R{\cdot}T$ the translation is applied first, so the offset itself gets rotated and scaled: the box ends up at $S\,R\,t = (-0.540, -0.059, 0.045)$, not at t. The scale also comes after the rotation, so it stretches along the *world* axes of an already rotated box. The first two columns are no longer orthogonal (their dot product is -0.092), and the box turns into a sheared parallelepiped. Figure 2 shows both cases. The wireframe "ghost" is always drawn with the order that is not in use. Everywhere else in the code a local transform is built as $T{\cdot}R{\cdot}S$.

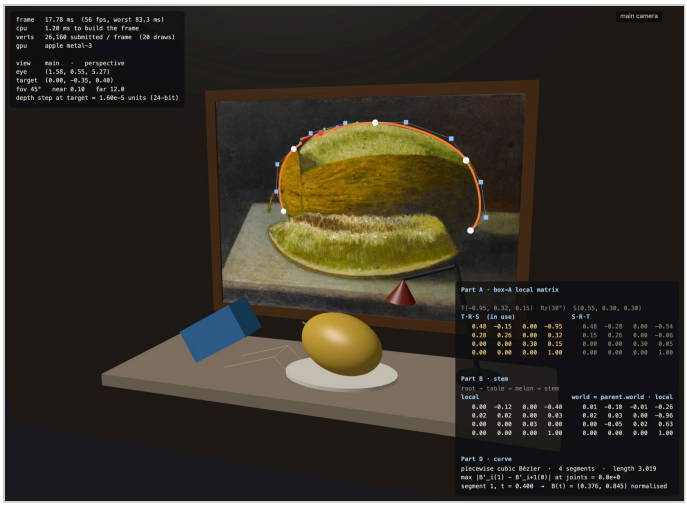


Figure 2a. $M = T{\cdot}R{\cdot}S$ (in use, highlighted). The yellow wireframe is the $S{\cdot}R{\cdot}T$ result.

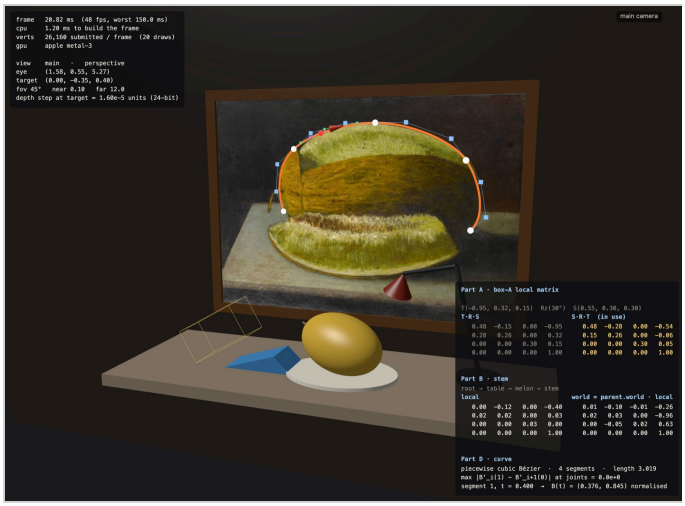


Figure 2b. $M = S{\cdot}R{\cdot}T$: the box moves towards the origin and is sheared by the non-uniform scale.

3 · Part B: scene graph

A `Node` holds a name, a local matrix, its parent and children, an optional mesh and an optional `update(node, time)` function. One depth-first traversal per frame computes $W_{\text{node}} = W_{\text{parent}} \cdot L_{\text{node}}$. Parents are always finished before their children. The graph has 25 nodes and 7 levels (Figure 3). Three joints are animated: the lamp's *swivel* turns about y by $0.45\sin(0.4t)$ rad, the *elbow* bends by $98° + 10°\sin(0.9t)$ and the *head* nods by $55° + 12°\sin(0.7t + 1)$. A rotation at the swivel therefore carries both arms, the shade and the bulb with it.

One design decision mattered here. Nodes that only position things (*table*, *melon*, *lamp*, *elbow*…) never carry a scale. The scaled geometry hangs below them as separate leaves (*melon-body*, *arm-1*…). Otherwise a non-uniform scale such as the melon's (0.78, 0.48, 0.50) would leak into the stem and squash it, the same shear effect as in Part A. The debug mode draws every node's local axes (Figure 4). The axes use a pipeline without depth test, so axes inside a mesh stay visible.

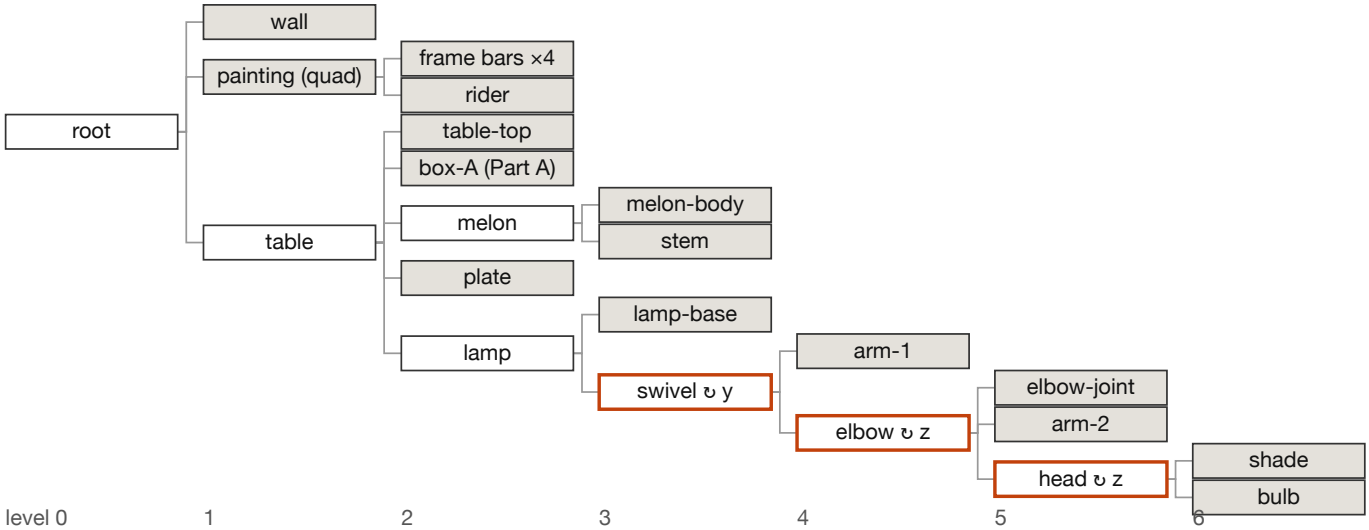


Figure 3. Scene graph. Grey boxes carry a mesh, white boxes are pure transforms, orange borders are animated joints. The painting's frame is drawn as one box here but is four nodes in the code.

World matrix of a leaf, checked by hand

The leaf *stem* has the path root → table → melon → stem, so $W = L_{\text{table}} \cdot L_{\text{melon}} \cdot L_{\text{stem}}$ with $L_{\text{table}} = T(0, -1.3, 0.75)$, $L_{\text{melon}} = T(0.1, 0.27, 0.05)\ R_y(-25°)\ R_z(-6°)$ and $L_{\text{stem}} = T(-0.4, 0.03, 0)\ R_z(80°)\ S(0.025, 0.12, 0.025)$. The stem's origin in world space is the last column of W:

- $R_z(-6°)\cdot(-0.4, 0.03, 0)$: $x = 0.99452(-0.4) + 0.10453(0.03) = -0.39467$, $y = 0.10453(0.4) + 0.99452(0.03) = 0.07165$.
- $R_y(-25°)$: $x = 0.90631(-0.39467) = -0.35769$, $z = -0.42262 \cdot 0.39467 = -0.16680$, y unchanged.
- Add the melon offset (0.1, 0.27, 0.05) and then the table offset (0, −1.3, 0.75): (−0.25769, −0.95835, 0.63320).

The program prints the following for *stem*. The translation column matches the hand result to four decimals. The 3×3 block is $R_y(-25°)R_z(-6°)R_z(80°)S$, whose columns have lengths 0.025, 0.12 and 0.025, as they should.

$$W_{\text{stem}} = \begin{pmatrix} 0.0062 & -0.1045 & -0.0106 & -0.2577 \\ 0.0240 & 0.0331 & 0 & -0.9584 \\ 0.0029 & -0.0487 & 0.0227 & 0.6332 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

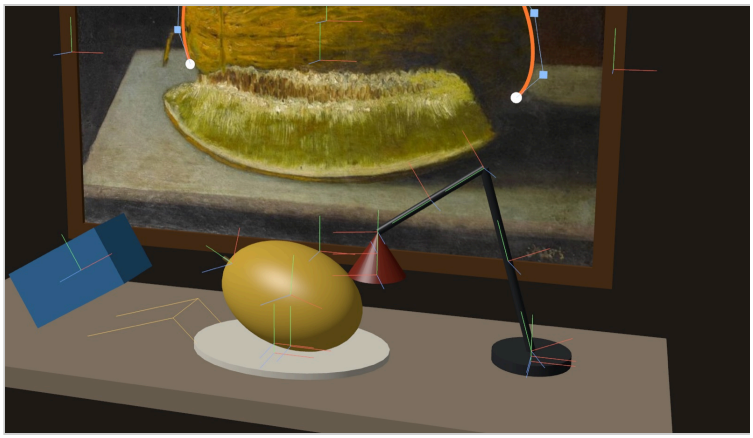


Figure 4. Debug mode: local axes of every node (x red, y green, z blue). Along the lamp each joint's frame is rotated relative to its parent.

4 · Part C: camera and projection

View matrix

The camera sits at e, looks at the target c, and, as usual, looks down its own $-z$ axis. Two cross products give an orthonormal frame:

$$w = (e - c) \,/\, \|e - c\|, \quad u = (up \times w) \,/\, \|up \times w\|, \quad v = w \times u.$$

The camera-to-world matrix is $C = [u\ v\ w\ e]$ (axes and origin as columns). This is a rigid transform, so the view matrix is its rigid inverse, $V = C^{-1} = [R^{\mathrm{T}} \mid -R^{\mathrm{T}}e]$. Its rows are u, v, w, and its translation column is $(-u{\cdot}e, -v{\cdot}e, -w{\cdot}e)$. The unit test checks that $V\,e = 0$ and that $V\,c$ lies on the negative z axis at distance $\|e - c\|$. **Degenerate case:** if up is parallel to w, then $up \times w = 0$ and u is undefined. When $\|up \times w\| < 10^{-6}$, `lookAt` switches to a different up vector: $(0, 0, -\text{sign}\ w_y)$ when the camera looks almost straight up or down, otherwise $(0, 1, 0)$. This choice keeps world x pointing right on screen in the overhead view. The *top view* button puts the camera exactly above the target to show this case (Figure 6b).

Perspective and orthographic projection for WebGPU

With $f = \cot(\text{fov}/2)$, x and y are scaled by f/aspect and f, and the divide by $w_{clip} = -z$ produces the perspective. For depth, write $z_{clip} = A\,z + B$. WebGPU wants $z_{ndc} = 0$ at $z = -n$ and 1 at $z = -f$:

$$(-An + B)/n = 0, \quad (-Af + B)/f = 1 \quad \Rightarrow \quad A = f/(n - f), \quad B = nf/(n - f).$$

OpenGL's $[-1, 1]$ version would have $A = (f + n)/(n - f)$ and $B = 2fn/(n - f)$. In WebGPU that matrix would send everything near the camera to $z_{ndc} < 0$, where it gets clipped. The orthographic matrix uses the same depth mapping, $z_{ndc} = (z + n)/(n - f)$. Its half-height is set to $d{\cdot}\tan(\text{fov}/2)$, where d is the orbit distance, so the target keeps its size when the projection is switched at run time (Figure 5). The mouse orbits (yaw/pitch on a sphere around the target), pans and zooms. Eye, target, fov, near and far are always shown on screen.

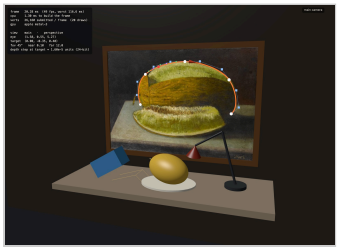


Figure 5a. Perspective, fov 45°, near 0.1, far 12.

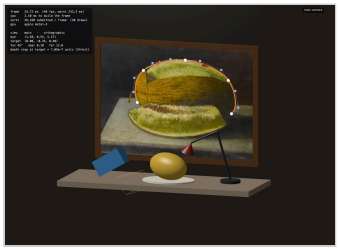


Figure 5b. Orthographic, same camera. Parallel edges of the table stay parallel.

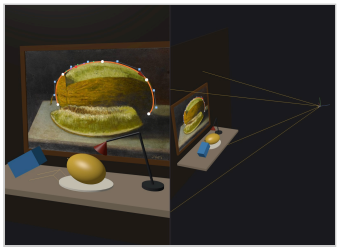


Figure 6a. Side by side: the inspection camera (right) sees the main camera's perspective frustum (yellow) and its axes.

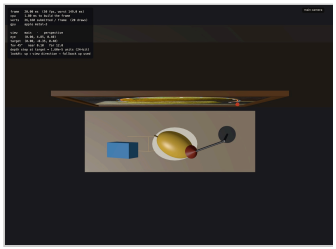


Figure 6b. Top view: *up* parallel to the view direction. The fallback up keeps x to the right, and the panel reports it.

The inspection camera is fixed and draws the main camera's frustum. Its corners come straight from the camera frame: at distance $d \in \{n, f\}$ the rectangle is centred at $e - d\,w$ with half-size $d\,\tan(\text{fov}/2){\cdot}(\text{aspect}, 1)$. For the orthographic case the half-size is constant. A unit test projects these corners back through $P\,V$ and gets NDC $(\pm 1, \pm 1, 0/1)$.

Near and far

The depth buffer stores z_{ndc}, which is proportional to $1/z$. Differentiating gives the smallest resolvable depth step at eye distance z, for a buffer with 2^{-24} resolution:

$$\Delta z \approx 2^{-24} \cdot z^2\,(f - n) \,/\, (n\,f) \approx 2^{-24}\,z^2 \,/\, n \quad (\text{for } f \gg n).$$

near	far	Δz at distance 5.2 (the target)	Δz at distance 10
0.1	12	$1.6{\cdot}10^{-5}$	$5.9{\cdot}10^{-5}$
0.1	100	$1.6{\cdot}10^{-5}$	$6.0{\cdot}10^{-5}$
0.01	12	$1.6{\cdot}10^{-4}$	$6.0{\cdot}10^{-4}$
0.001	12	$1.6{\cdot}10^{-3}$	$6.0{\cdot}10^{-3}$

The far plane hardly matters, but every factor of ten taken off the near plane costs a factor of ten in precision. I use **near = 0.1, far = 12**. The orbit distance is clamped to at least 1.2, so in normal use the geometry stays well beyond the near plane. The whole scene (wall at about 5–6 units from the default eye) fits inside 12 with margin, and a short far plane keeps the frustum drawing readable. The curve is lifted 0.004–0.01 units off the painting. That is more than 100 times the depth step at this distance, so it does not z-fight with the quad. With near = 0.001 the step would be $1.6{\cdot}10^{-3}$, and the ribbon would be only two or three depth steps in front of the painting, close enough to start fighting with it. The panel shows the current Δz live.

5 · Part D: curves over the painting

Bézier basis. A cubic segment is $B(t) = \Sigma\, P_i\, b_i(t)$ with the Bernstein polynomials $b_0 = (1-t)^3$, $b_1 = 3t(1-t)^2$, $b_2 = 3t^2(1-t)$, $b_3 = t^3$, which sum to $(t + 1 - t)^3 = 1$.

Endpoints. At $t = 0$, $b_0(0) = 1$, and b_1, b_2, b_3 all contain the factor t, so they vanish and $B(0) = P_0$. At $t = 1$, $b_3(1) = 1$, and b_0, b_1, b_2 contain the factor $(1 - t)$, so $B(1) = P_3$. ∎ Differentiating, $B'(t) = 3[(1-t)^2(P_1 - P_0) + 2t(1-t)(P_2 - P_1) + t^2(P_3 - P_2)]$, so $B'(0) = 3(P_1 - P_0)$ and $B'(1) = 3(P_3 - P_2)$. The handles set the end tangents.

C¹ at a joint. The piecewise curve is stored as $P_0 \ldots P_{3n}$. Segment k uses $P_{3k} \ldots P_{3k+3}$, so two segments share the knot $K = P_{3k}$. Position continuity (C⁰) holds by construction. C¹ needs $3(K - P_{3k-1}) = 3(P_{3k+1} - K)$, i.e.

$$P_{3k+1} = 2K - P_{3k-1} \quad \text{(the two handles are mirror images through the knot).}$$

The editor keeps this true. Dragging a handle moves the mirrored handle on the other side, and dragging a knot moves both of its handles by the same amount. The panel prints $\max \|B'_k(1) - B'_{k+1}(0)\|$ over all joints, which stays at 0 up to rounding (about 10^{-16}) after any edit.

de Casteljau. For a chosen segment and t the app draws the two intermediate levels of linear interpolation (three green points, then two purple ones) and the final point (red), which is $B(t)$ (Figure 7b). A test checks that it agrees with the Bernstein form.
Catmull–Rom. As the second curve family I implemented the uniform Catmull–Rom spline through the knots, $C(t) = \tfrac{1}{2}[2P_i + (P_{i+1} - P_{i-1})t + (2P_{i-1} - 5P_i + 4P_{i+1} - P_{i+2})t^2 + (-P_{i-1} + 3P_i - 3P_{i+1} + P_{i+2})t^3]$. It passes through every point with tangent $(P_{i+1} - P_{i-1})/2$. Each segment is the Bézier with handles $P_i + (P_{i+1} - P_{i-1})/6$ and $P_{i+1} - (P_{i+2} - P_i)/6$, which a unit test confirms. The end segments use reflected ghost points $(2P_0 - P_1)$. Both curve types share the same knots, so I can switch between them while tracing (Figure 7c).

Tracing and interaction. The control points live in the painting's plane. A mouse press is turned into a ray from the camera, the ray is intersected with the plane of the painting node, and the hit is mapped into the painting's frame with its rigid inverse. Picking happens in screen space (12 px radius), so it works the same under both projections. Double-clicking adds a segment. I placed the knots on the outline of the whole melon: the bottom of the cut face, the top of the cut, the top of the rind, the shoulder and the right end. I then moved the handles until the middle of each segment lay on the contour. **Animation.** The rider moves at constant speed: a 400-sample arc-length table maps distance to parameter by binary search. Its frame is $[T \times Z, T, Z, p]$, with T the unit tangent and Z the painting normal, so the cone always points along the curve. **Export** writes `curve.json` with normalised coordinates, (0, 0) at the bottom-left and (1, 1) at the top-right (Appendix A).

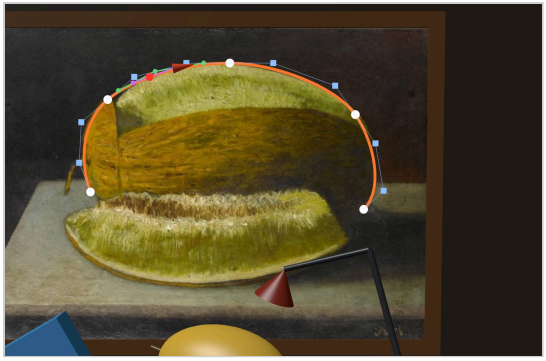


Figure 7a. Traced outline of the melon: 4 cubic Bézier segments, C¹ at the joints. White discs are knots, blue squares are handles. The red cone rides the curve.



Figure 7b. de Casteljau construction on segment 2 at $t = 0.5$. The red point is $B(0.5)$.



Figure 7c. Catmull–Rom spline through the same knots.

6 · Part E: integration and performance

Everything is one page. The code is split into `math/` (vectors, matrices, curves: no WebGPU), `scene/` (nodes, the scene, cameras, the editable curve), `render/` (device setup, WGSL, pipelines, procedural meshes) and `ui/` (panel, mouse, text overlays). The renderer has four pipelines: lit meshes, the textured painting, lines and flat overlay triangles, plus the overlay-line variant for the axes. Per-object and per-view uniforms live in two buffers addressed with 256-byte dynamic offsets, so one bind group serves every draw. Rendering uses 4× MSAA. The painting is an `rgba8unorm-srgb` texture and the frame is rendered through an `-srgb` view of the canvas, so lighting is computed in linear space and encoded once on output. The controls switch views and projection, reset the camera, the curve or the whole scene, and toggle every debug drawing. 26 unit tests (`npm test`) cover the maths, the scene graph and the camera.

Mode (canvas 2600×1900 px, 2× DPR)	Frame time avg / p95 / max	CPU per frame (median / max)	Vertices / frame	Draws
Main camera	16.67 / 17.7 / 17.8 ms	1.9 / 2.9 ms	26 190	20
Side by side	16.67 / 17.7 / 17.8 ms	1.8 / 3.2 ms	52 418	40
Side by side + node axes	16.95 / 17.7 / 33.4 ms	1.9 / 4.3 ms	52 706	42

Measured on an **Apple M3 (10-core GPU), macOS 26.3.1**, in a Chromium 152 based browser, over 5 s (about 300 frames) per mode. "Vertices" counts indices for indexed draws. The frame time sits at the 60 Hz refresh interval in every mode, so the GPU is not the limit for this scene. The one 33 ms frame is a single missed refresh. CPU time (animation, world matrices, line and triangle batches, uniform uploads) is about 2 ms. I did not use GPU timestamp queries, so the pure GPU time is not measured. The text panels are refreshed ten times a second rather than every frame, because their content does not need to change faster than that.

7 · What went wrong and how I found it

- **The inspection camera "could not see" the main camera.** The yellow frustum lines ran off the right edge of the view. Projecting the main camera's eye by hand gave $x_{ndc} \approx 0.67$, so it should have been visible. The real cause was the control panel, which was drawn on top of the canvas and covered exactly that region. The panel now has its own column and the canvas ends where the panel begins.
- **The first trace was off at the melon's right end.** Viewed head-on, the knot I had put at (620, 130) px was outside the outline. At that height the contour is at about $x = 597$. I enlarged crops of the image, read off points on the outline, and replaced the knots so that each segment's midpoint ($t = ½$, easy to compute by hand) falls on the contour.
- **The overhead view came out rotated by 90°.** My first fallback in `lookAt` took the world axis least aligned with w as the new up vector. For a camera looking straight down that is the x axis, so the screen's up pointed along world x and the scene appeared on its side. I noticed this while working out by hand what the top view should look like. The fallback now uses the z axis for near-vertical views, and a unit test checks that world $+x$ still maps to screen right.
- **The debug axes were invisible.** The origins of most nodes are inside their own meshes, so the depth test hid the axes. Drawing them in a separate pass with `depthCompare: 'always'` fixed this without affecting the curve overlays, which must stay depth-tested.

Appendix A · Exported curve (`public/curve.json`)


```
{ "painting": "Kavunlu Natürmort", "artist": "Şeker Ahmed Paşa",
  "coordinates": "normalised to the painting: (0,0) bottom-left, (1,1) top-right",
  "type": "bezier", "continuity": "C1 (handles mirrored at every joint)", "segments": 4,
  "points": [
    { "x": 0.2417, "y": 0.4781, "role": "knot" },    { "x": 0.2167, "y": 0.5717, "role": "handle" },
    { "x": 0.2222, "y": 0.7012, "role": "handle" },  { "x": 0.2819, "y": 0.7729, "role": "knot" },
    { "x": 0.3417, "y": 0.8446, "role": "handle" },  { "x": 0.4583, "y": 0.8884, "role": "handle" },
    { "x": 0.5556, "y": 0.8884, "role": "knot" },    { "x": 0.6528, "y": 0.8884, "role": "handle" },
    { "x": 0.7917, "y": 0.8167, "role": "handle" },  { "x": 0.8375, "y": 0.7251, "role": "knot" },
    { "x": 0.8833, "y": 0.6335, "role": "handle" },  { "x": 0.9028, "y": 0.4821, "role": "handle" },
    { "x": 0.8583, "y": 0.4223, "role": "knot" } ] }
```


The file in the repository has one key per line; it is reformatted here to save space. Check C^1 at the second knot: $2 \cdot (0.2819, 0.7729) - (0.2222, 0.7012) = (0.3416, 0.8446)$, which is the next handle (to rounding).

Appendix B · Use of AI tools

Claude Opus 5.5 was used to generate test scripts and debug some of the code. Opus 5.5 also helped with generating the report based on the app.